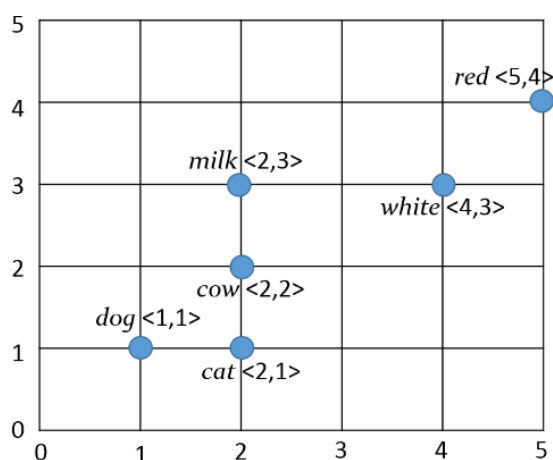


1. Word Embeddings

Problem designed by Monojit Choudhury

Computers can “compute” mathematical functions that take numbers as input and give numbers as output. Therefore, in order to make computers understand and process human language, it is useful to represent words in terms of numbers. Computer scientists have come up with sophisticated techniques that encode words as “vectors”, i.e., a sequence of numbers. These are called *word embeddings*. It might be easier to think of word embedding as a mapping of words to points in an n -dimensional space. Let us assume that $n = 2$, i.e., words are mapped to points in a two-dimensional plane.



The diagram in the left shows hypothetical word embeddings for 6 English words. As you can see in this example, words which have similar meanings or are used in similar context have word embeddings which are close to each other, i.e., the points are closer in space. Note: The embeddings in this example are NOT related to the problem below.

Here are some English words in alphabetical order:

boy, first, girl, grammar, language,
literature, man, mathematics,
mathematician, number, one, position,
queen, second, time, two, woman

Shown below, in no particular order, are the translations of these words in the Santali language written in the Ol Chiki script, and their (the English words’) embeddings.

#	Santali Translation	English Embedding
A	ᱵᱤᱨᱫᱟ	<3,3>
B	ᱵᱤᱨᱫᱟᱵᱟᱫᱽ	<3,2>
C	ᱵᱤᱨᱫᱟ	<5,4>
D	ᱵᱟᱦᱟ	<7,2>
E	ᱵᱟᱨᱫᱟ	<7,4>
F	ᱵᱟᱦᱟᱵᱟᱫᱽ	<8,2>
G	ᱵᱟᱨᱫᱟᱵᱟᱫᱽ	<8,4>
H	ᱵᱟᱨᱫᱟ	<9,3>
I	ᱵᱟᱨᱫᱟ	<10,6>

#	Santali Translation	English Embedding
J	ᱵᱟᱨᱫᱟ	<8,13>
K	ᱵᱟᱨᱫᱟ	<4,7>
L	ᱵᱟᱨᱫᱟ	<5,6>
M	ᱵᱟᱨᱫᱟ	<3,8>
N	ᱵᱟᱨᱫᱟ	<9,10>
O	ᱵᱟᱨᱫᱟ ᱵᱟᱨᱫᱟ	<11,11>
P	ᱵᱟᱨᱫᱟ	<8,12>
Q	ᱵᱟᱨᱫᱟ ᱵᱟᱨᱫᱟ	<10,13>

Assignment 1 [Answer Booklet]: Match the English words to their Santali translations – A to Q. No explanation needed.

Assignment 2 [Answer Booklet]: Can you guess the embedding of the words: *king*, *linguist*, *princess*? Explain your answer.

Assignment 3 [Answer Booklet]: Give the English translation of the following Santali word.

ᱢᱟᱞᱟᱝ ᱢᱟᱞᱟᱝ ᱢᱟᱞᱟᱝ.

Notes: Santali is a language in the Munda subfamily of Austroasiatic languages. It is spoken by around 6.2 million people in India, Bangladesh, Bhutan and Nepal. Most of its speakers live in India, in the states of Jharkhand, Bihar, Odisha, Tripura, Mizoram, Assam and West Bengal. Since Santali did not have a script, it was written using the Roman script or the Eastern Nagari (Bangla) script. However, none of the existing scripts were able to phonetically represent the Santali language. This resulted in the invention of a new script called *Ol Chiki* by Pandit Raghunath Murmu in 1925. He is popularly known as Guru Gomke among the Santals.